Meihui (Lily) Wu

Los Angeles, CA | (323) 639-6232 | meihui027@outlook.com | LinkedIn | GitHub | Portfolio

Education

University of Southern California — Los Angeles, CA
B.S. Applied & Computational Mathematics; B.A. Cognitive Science — *Expected May 2027*

- **GPA:** 3.88 | **Selected Coursework:** Neural & Cognitive Data Science, Statistical Consulting & Data Analysis (graduate), Neural Network Models of Social and Cognitive Processes, Mathematical Statistics

Research Interests

Conversation dynamics and turn-taking; multimodal behavioral signals (voice, language, timing); computational approaches to social connection and mental health

Publications & Preprints

Zhou, C., **Wu, M.**, Xiang, Y.-M., & Itti, L. (2026). *Cross-Modal Benchmarking of Acoustic Prosody and Ventral Striatal BOLD for Depression-Related Anhedonia Classification: A Pre-Registered Study with the ClinicalWhisper Pipeline.* bioRxiv preprint. doi:10.64898/2026.06.08.728970

Research Experience

Undergraduate Research Assistant — Jan. 2025 – Present
USC Social Connection Lab (PI: Elisa Baek) — *Los Angeles, CA*

- **Showed that turn-taking pace was the strongest predictor of social connection** in 46 stranger conversations ($r = -.70$), and replicated the effect in 1,603 conversations from the CANDOR corpus.
- Sole developer of the study's end-to-end pipeline (transcription, speaker separation, acoustic and pose extraction), converting 46 video sessions into features for 5,425 speaking turns.
- Applied a reliability-first analysis, testing each feature's cross-session stability before relating it to outcomes; vocal synchrony and body movement did not predict connection.
- Quantified linguistic alignment, emotional tone, and 10 interaction behaviors with LLM annotations validated against human coders; **first-authored a SANS 2026 poster**.
- **Awarded two PURF fellowships** for this line of work: multimodal modeling of the co-viewing study (Summer 2026) and a project on how speaking-time balance shapes partners' beliefs about being liked (Fall 2026).
- As lead research coordinator, ran **20+ dyadic fNIRS sessions**, from participant setup to signal checks.

Audio Stream Lead, ClinicalWhisper — Jan. 2026 – Present
Student-led independent research, USC (Faculty Advisor: Laurent Itti) — *Los Angeles, CA*

- Student-initiated project of a three-person core team developing speech-based markers of anhedonia (loss of interest or pleasure), which is usually measured only by self-report; as audio stream lead, built the speech pipeline (transcription, diarization, 32 vocal and timing measures).
- **Lead a study of 2,581 answers from 135 clinical interviews** with a virtual interviewer (DAIC-WOZ), testing whether anhedonia weakens how the voice tracks positive language; abstract for WPA 2027.
- Found that vocal arousal rose with positive content ($b = 0.06$, $p = .001$) but was not clearly moderated by anhedonia; person-level responses were unreliable, and increased pausing emerged as an exploratory marker.
- Validated LLM positivity ratings against two human raters (ICC = .62–.65); tested 32 vocal and timing measures, controlling for low mood and psychomotor slowing with multiple-comparison correction, and used filtered speech to separate words from voice.
- **Co-authored a preregistered bioRxiv preprint** benchmarking acoustic prosody against ventral striatal BOLD for anhedonia classification (AUC = 0.63).
- Recruited **five student researchers** through Praxis, USC's computational psychiatry organization, to expand the team this semester; lead a two-person audio subgroup with one recruit.

Undergraduate Research Assistant — Sep. 2026 – Present
USC Intelligent Human Perception Lab (PI: Mohammad Soleymani) — *Los Angeles, CA*

- Investigate how voice, language, and turn-taking reveal trust, dominance, and deception in multiparty social-deduction games, curating 800+ hours of multi-camera video from 82 games across seven international sites.
- Built an automated audit-and-repair pipeline for a legacy corpus, **increasing verified recordings from 55% to 95.6%** (3,283 segments).
- Developed a multimodal speaker diarization model for nine-person conversations, **reducing diarization error to 10.7%**, 7 points below a pretrained pyannote baseline.

Additional Research Experience

Undergraduate Research Assistant Sep. 2024 – Dec. 2025
USC Neural Modeling and Interface Lab (PI: Dong Song) *Los Angeles, CA*
- Developed Python and MATLAB tools to convert raw hippocampal Neuropixels recordings for Kilosort4; spike-sorted and analyzed single-unit activity from TMS experiments.

Summer Research Group Member Jun. 2025 – Aug. 2025
USC Neural Vitality Lab (PI: Bruna Martins-Klein) *Los Angeles, CA*
- Preprocessed and quality-checked 750+ MRI scans on HPC, correcting signal dropout and registration errors to support cross-site reproducibility.

Course Projects

Instagram Post Virality Prediction (MATH 446) *Co-author • Fall 2025*
- Framed virality as rare-event classification on 22,924 posts; similarity-based sample filtering with a gradient-boosting ensemble **raised F1 from 0.44 to 0.58**.

Kin: Care Coordination App (TAC 310) *Team Lead • Spring 2026*
- Led user research for a patient–caregiver care coordination app, using competitive analysis, journey mapping, and prototype testing to guide multiple design iterations.

Conference & Research Presentations

Wu, M., Lu, C., Grady, S. M., & Baek, E. C. (2026). *Linguistic Alignment and Conversation Dynamics as Predictors of Social Connection in Dyadic Co-viewing.* Poster presented at the Social & Affective Neuroscience Society Annual Conference.

Zhou, C., **Wu, M.**, Xiang, Y.-M., & Itti, L. (2026). *ClinicalWhisper: Cross-Modal Benchmarking of Acoustic Prosody and Ventral Striatal BOLD for Depression-Related Anhedonia Classification.* Poster presented at the USC Undergraduate Research Symposium.

Honors & Research Funding

- USC Provost's Undergraduate Research Fellowship (PURF) Summer 2025, Summer 2026, Fall 2026
- Jessie Ingram Endowed Scholarship, USC 2026–2027
- Lenore B. Kreiger Endowed Scholarship for Mathematics, USC 2025–2026
- Cathay Bank Scholarship (one of six Southern California recipients) 2025
- Sylas and Rose Marx Meyer Scholarship, USC 2024–2025

Leadership

Director of Education & Project Lead Sep. 2024 – Present
NeuroTech@USC (Director of Education since Sep. 2026) *Los Angeles, CA*
- Designed and teach the weekly NeuroTech Academy, **6+ one-hour sessions** with worksheets that bridge **40 beginners** across majors from neurotechnology basics to building their own projects.
- Built the Academy's learning portal so members can track progress and review concepts between sessions.
- As project lead, guided a member team whose eye-tracking study of bilingual code-switching was presented to **400+ attendees** at the 2025 California Neurotechnology Conference.

Vice President Sep. 2026 – Present
Praxis, USC Computational Psychiatry Organization *Los Angeles, CA*
- Founding officer; run the biweekly computational psychiatry journal club and coordinate weekly research updates across member projects.
- Turned the club into a research pipeline, **recruiting five members** into the ClinicalWhisper project this semester.
- Produce the Praxis podcast from invited-speaker talks, handling recording, transcripts, and episode materials.

Technical & Research Skills

Programming	Python (PyTorch, scikit-learn, pandas), MATLAB, C++, HPC/GPU computing
Statistics & ML	Mixed-effects models, permutation and bootstrap inference, reliability analysis, ensembles
Multimodal	Speech (Whisper, pyannote, eGeMAPS), text (SBERT, LLMs), video (MediaPipe, OpenCV)
Neural Data	fNIRS, fMRI, EEG, Neuropixels